Does history help, and does the path signature help?

An a-posteriori benchmark of value-gradient representations for continuous-time control of delayed, non-Markovian dynamics

Benchmark report — `RL_and_signatures`

2026-06-30

Abstract

Three feature maps for the value function of a Doya-style continuous-time value-gradient controller are benchmarked on a graded family of delayed (non-Markovian) plants, by two complementary instruments: a comprehensive *learned* study sweeping representation capacity (polynomial degree, signature depth) across many examples and five seeds, and a *controlled* continuous-time optimal-control oracle protocol that isolates the representational question by solving the fit exactly. (The originally-intended nonlinear-delayed example, a continuous-stirred-tank reactor, is measured to be structurally near-Markovian and is demoted; a delay-coupled Hopfield network replaces it.) Two hypotheses are tested by the closed-loop cost J: H_1, that a representation of the *history* window improves on a representation of the current state alone; and H_2, that the *path-signature* structure improves on a plain polynomial of the history window. The first hypothesis **holds** wherever the delay genuinely couples the dynamics — the linear delay equation, the delayed Hopfield network, the five-vehicle platoon, and the Mackey–Glass oscillator — and fails exactly on the Markovian negative control, where all three representations coincide at $J = 3.115$ (measured, five seeds). The second hypothesis does *not* hold. Early single-seed evidence that the signature beats the history window on long-delay examples is shown to be an artefact of *under-sampling*: the degree-two history feature dimension grows as $O(\tau^2)$ with the delay τ (window $6 \mapsto 61$, dimension $90 \mapsto 7625$), so on a long-delay example at fixed sample size the history fit is under-sampled. Under three controls — quadrupled data, Tikhonov conditioning, and off-sheet rebalancing — the history representation recovers and *matches or beats* the signature on every example measured (Hopfield at $\tau = 3$: $J = 0.289$ versus 0.323; Mackey–Glass limit cycle at off/dim = 4: $J = 0.0108$ versus 0.0126). The mechanism is the effective-rank phenomenon of the companion note, proven as a Veronese bound and measured here: the controlled-trajectory feature Gram has an effective rank that rises only from 3.3 to 3.5 across $\tau \in \{1, \ldots, 6\}$ while the nominal dimension increases by a factor of 28. The signature's sole genuine advantage is therefore *sample efficiency* — a fixed feature dimension — not representational power; and that advantage reverses at high state-space dimension, where the signature's own $O(d^L)$ channel growth makes it the ill-conditioned representation (platoon: $J = 0.227$ versus the history window's 0.029). Every number is read off a saved run; the reproduction appendix lists the script and run directory behind each.

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1 Common framework

This section fixes, once, the objects shared by every test case: the control problem, the oracle that plays the reference, the representations under test, the learning problem and its data protocol, and the three metrics. The per-example sections that follow vary only the plant.

1.1 The control problem and Doya's value-gradient law

Definition 1.1 (Controlled delay system and cost)**.** A plant is a controlled delay differential equation on $\mathbb{R}^n$,

$$\dot{x}(t) = f\big(x(t),\, x(t-\tau),\, u(t)\big), \qquad x(s) = x_0(s) \quad (s \in [-\tau, 0]), \tag{1}$$

with control $u(t) \in \mathbb{R}^m$ and a quadratic running cost $\ell(x,u) = x^\top Qx + u^\top Ru$, $Q \succeq 0$, $R \succ 0$, regulating to the origin (a shifted equilibrium). The closed-loop performance of a feedback law is the accumulated cost

$$J = \int_0^T \ell\big(x(t), u(t)\big)\, \mathrm{d}t \tag{2}$$

from a fixed deployment initial condition, integrated by the environment's own Runge–Kutta solver. J is the operative quantity: every verdict in this report is a comparison of J.

Definition 1.2 (Value-gradient feedback)**.** Let J^* denote the optimal cost-to-go of (1)–(2) — the minimal accumulated cost (2) attainable from the current history — and let the value be its negative, $V^* := -J^*$ (reward convention). Doya's continuous-time law sets the control to the half-gradient of the value along the input channel,

$$u = \tfrac{1}{2}\, R^{-1} B^\top\, \partial_x V^*, \qquad B := \partial_u f, \tag{3}$$

which reproduces the optimal control exactly for the continuous-time value (no time-step rescaling). A learned critic V_θ is deployed by the same law, $u_\theta = \frac{1}{2} R^{-1} B^\top \partial_x V_\theta$, so what is *used* is the *gradient* $\partial_x V_\theta$, not V_θ itself — a distinction that drives the entire analysis (Section 6).

1.2 The reference: the continuous-time oracle and its consistency check

Definition 1.3 (Continuous-time oracle). The reference controller is the continuous-time optimal control of (1), built per example.

- *Linear / linearised examples (delayed-LQR).* The history segment $[-\tau, 0]$ is discretised at the $\nu+1$ Chebyshev collocation nodes $-\tau = s_\nu < \cdots < s_0 = 0$, and the augmented state $z = \big(x(t+s_0), \ldots, x(t+s_\nu)\big) \in \mathbb{R}^{n(\nu+1)}$ stacks the state at those nodes. The delay equation (1) becomes the ordinary differential equation $\dot{z} = Mz + \mathcal{N}u$, in which $M \in \mathbb{R}^{n(\nu+1)\times n(\nu+1)}$ collocates the transport identity $\partial_t x = \partial_s x$ at the interior nodes and the linearised dynamics $\dot{x} = A_0\, x(t) + A_1\, x(t-\tau)$ at the boundary node $s_0 = 0$, and $\mathcal{N} = (B^\top, 0, \ldots, 0)^\top \in \mathbb{R}^{n(\nu+1)\times m}$ injects the control at that node. The algebraic Riccati equation $M^\top P_c + P_c M - P_c \mathcal{N} R^{-1} \mathcal{N}^\top P_c + Q_z = 0$, with Q_z equal to Q on the boundary block and 0 elsewhere, yields the value $V^*(z) = -z^\top P_c z$ and the feedback $u^* = -K_c z$ with $K_c = R^{-1}\mathcal{N}^\top P_c$. (`src/solvers/continuous_oracle.py`.)
- *Nonlinear examples (Pontryagin).* On the same augmented state the two-point boundary-value problem in (z, λ) is solved, with λ the costate; by the maximum principle $\lambda(t) = \partial_z V^*$ is the value gradient, so $u^* = -\frac{1}{2} R^{-1} B^\top \lambda_0$ (with λ_0 the boundary block of λ) is Doya's law (3) evaluated on the exact value. (`src/solvers/{mackey_glass,delayed_hopfield}_pontryagin.py`.)

Consistency check. The oracle is accepted only after a numerical consistency check: on a cloud of augmented states, Doya's law (3) applied to the oracle value gradient must reproduce the oracle feedback u^* to machine precision — direction cosine 1 and magnitude ratio $\|u\| \,/\, \|u^*\| = 1$. Every example in this report passes this check to four decimals (recorded as the `gate` field of each `summary.json`); an example whose oracle failed it would be excluded.

The oracle is a deliberately strong, non-deployable adversary: it has the exact dynamics and solves a boundary-value problem per window. It plays two roles — it *labels* the training data with (V^*, u^*), and it supplies the deployable linear-feedback reference cost J^* against which the learned J is read.

1.3 The two hypotheses

With the three representations of Definition 1.4 ranked markovian → raw-history → signature, the two questions under test are the adjacent pairwise comparisons of the closed-loop cost:

- H_1 (*does history help?*): J[raw-history] < J[markovian];
- H_2 (*does the signature structure help?*): J[signature] < J[raw-history].

Each is recorded as holding on an example only if the contender lowers J by more than a relative margin of 3%; a near-tie is the negative result. The single-example verdict is indicative; the falsifiable statement is the seed-resolved comparison of Section 4.

1.4 The representations under test

Definition 1.4 (Markovian, raw-history, signature critics). The control-cadence history at time t is the finite sequence $\underline{x} = (x_{t-i\Delta t})_{0\le i\le L-1} \in (\mathbb{R}^n)^L$ of the last L state samples at the control step Δt — x_t the current state and $x_{t-(L-1)\Delta t}$ the oldest — with $L = \lceil \tau/\Delta t \rceil + 1$ so the sequence spans the delay. The value critic is linear in a feature map ϕ, $V_\theta(\underline{x}) = \langle \theta,\, \phi(\underline{x}) \rangle$, with the three maps

1. *markovian* ϕ_M: the degree-2 monomials of the *current* state x_t alone (no history);
2. *raw-history* ϕ_H: the degree-2 monomials of the *whole* sequence $\underline{x}$ (its Ln scalar entries);

3. *signature* ϕ_S: the depth-2 path signature of the time-augmented sequence $\big(i\Delta t,\ x_{t-i\Delta t}\big)_{0\le i\le L-1}$.

ϕ_M is the H_1 null (a state-only critic); ϕ_H adds the history linearly in the sequence; ϕ_S adds the nonlinear path-functional structure. Their feature dimensions are tabulated per example in Section 2; the contrast in how those dimensions grow is the subject of Section 6.

1.5 The learning problem and its data protocol

Definition 1.5 (Data: on-sheet and off-sheet)**.** The training set $\{(\underline{x}_k,\ V^*(\underline{x}_k))\}_k$ is generated by one procedure, uniform across examples (`run/study/h1h2_evaluation.py`), from two families.

- *On-sheet.* A cloud of n_{ic} initial histories $\{x_0^{(j)}\}_{j=1}^{n_{\mathrm{ic}}}$ is drawn; from each, the optimal (oracle) trajectory $t\mapsto x^*(t;x_0^{(j)})$ is integrated, and its control-cadence sequences are collected along it: for a grid of times t_ℓ,

$$\underline{x}^{(j,\ell)} = \big(x^*(t_\ell - i\Delta t;\ x_0^{(j)})\big)_{0\le i\le L-1}, \qquad \text{labelled by}\quad V^*\big(\underline{x}^{(j,\ell)}\big).$$

  These are the states a deployed controller actually visits; there are n_{on} of them.

- *Off-sheet.* A sequence $\underline{x}$ is drawn uniformly from the on-sheet family and displaced,

$$\underline{x}' = \underline{x} + \varepsilon\,\Xi, \qquad \Xi = (\Xi_i)_{0\le i\le L-1} \text{ with i.i.d. } \Xi_i \sim \mathcal{N}(0,\mathrm{Id}), \quad \varepsilon = 0.25$$

  (the state is of order unity in every example). The perturbed sequence is *re-labelled through the oracle*: it is mapped to the augmented state $z(\underline{x}')$ by interpolating its samples onto the collocation nodes $s_0,\dots,s_\nu$, and $V^*(\underline{x}')$ is read from the oracle at $z(\underline{x}')$ — in closed form ($-z^\top P_c z$) for the linear examples, by one Pontryagin boundary-value problem for the nonlinear examples. There are n_{off} such sequences.

The off-sheet sequences carry the value *off* the optimal manifold, hence constrain the value gradient transverse to the trajectory — the quantity (3) deploys and the object of Section 6. The sample size $n_{\mathrm{train}} = n_{\mathrm{on}} + n_{\mathrm{off}} > \dim\phi$ in every reported fit; the ratios $n_{\mathrm{train}}/\dim\phi$ and $n_{\mathrm{off}}/\dim\phi$ are reported because they are the binding quantities for H_2.

Definition 1.6 (The two estimators)**.** Let $\Phi \in \mathbb{R}^{n_{\mathrm{train}}\times\dim\phi}$ be the design matrix with rows $\phi(\underline{x}_k)^\top$, let $y\in\mathbb{R}^{n_{\mathrm{train}}}$ collect the oracle labels $y_k = V^*(\underline{x}_k)$, and let $G = \Phi^\top\Phi$ be the (uncentred) empirical Gram. Two estimators of θ are used.

- *Oracle half* (the primary evidence). The regularised normal-equation solution

$$\hat\theta = \Big(G + \lambda\,\tfrac{\operatorname{tr} G}{\dim\phi}\,\mathrm{Id}\Big)^{-1}\Phi^\top y,$$

  with scale-aware penalty $\lambda \ge 0$ (the parameter `H1H2_RIDGE`); at $\lambda = 0$ this is the minimum-norm least-squares solution $\hat\theta = \Phi^+ y$. Because the labels are the *exact* oracle values, this estimator isolates the representation from the learning dynamics.

- *Learning half* (the confirmation, Appendix B). Damped least-squares policy iteration. At iteration k the policy π_k is frozen; on-policy transitions $(\underline{x},\underline{x}',r)$ are collected under π_k with additive exploration; the least-squares temporal-difference solution θ_k^{LSTD} of the projected Bellman equation is formed (the temporal-difference analogue of the oracle-half normal equations, its discount set by a gentleness parameter for the near-critical plants); the update is damped, $\theta_{k+1} = (1-\alpha)\theta_k + \alpha\,\theta_k^{\mathrm{LSTD}}$; and the policy is improved greedily by the value-gradient law (3). The lowest-cost iterate is restored.

1.6 The three metrics

Each fit is scored on three axes, one symbol each, on the deployment trajectory:

- value-fit $R^2 \in (-\infty, 1]$ — *expressivity*: the coefficient of determination of V_θ against V^* on the training sequences;
- gradient cosine $\eta := \dfrac{\langle u_\theta,\, u^* \rangle}{\|u_\theta\|\,\|u^*\|} \in [-1, 1]$, the cosine of the angle between the deployed control u_θ and the optimal control u^* stacked along the deployment trajectory — *extraction*: the alignment of the deployed value-gradient control with the optimal control along the deployment trajectory;
- closed-loop cost J (Equation (2)) — *control*: the operative quantity of comparison, read against the no-control cost J^0 and the oracle-feedback cost J^*.

The three are ordered by stringency: $R^2 \to 1$ is necessary but far from sufficient for $\eta \to 1$, which is in turn necessary but not sufficient for $J \to J^*$. The gap between $R^2 = 1$ and a large J is the value/gradient separation of Section 6.

2 Catalogue of the controlled-protocol examples

This is the roster of the controlled-oracle protocol of Sections 4–5, complementary to the comprehensive learned examples of Section 3 (`linear_cell`, `delayed_velocity`, `delayed_oscillator`, `strongly_delayed`, `dadebo`). The eight examples here populate a falsifiability grid: from a Markovian negative control (where H_1 must fail), through linear delay examples (where H_1 holds and H_2 is a provable negative), to nonlinear delay examples (where H_2 could in principle hold). The delay-coupled Hopfield network is the nonlinear-delayed primary example that replaces the demoted Dadebo example (Remark 3.2). Each is read from `make_cell` in `run/study/h1h2_evaluation.py`; the dynamics, the regulated equilibrium, and the oracle type are stated once here.

Definition 2.1 (The examples)**.**

- **markovian** (negative control). The double integrator $\ddot{x} = u$ on $\mathbb{R}^2$, no delay; V^* depends on $x(t)$ alone, so history is irrelevant by construction. Oracle: the standard continuous algebraic Riccati equation.
- **linear_dde**. The scalar linear delay equation $\dot{x}(t) = -1.5\,x(t-\tau) + u$, $\tau = 1$; V^* is exactly quadratic in the augmented state, so the history window spans it and H_2 is a provable negative. Oracle: delayed-LQR.
- **hopfield_linear**. The delay-coupled network $\dot{x} = -x + Wx(t-\tau) + u$ on $\mathbb{R}^2$ with a scaled rotation W (eigenvalues $2e^{\pm i\pi/2}$), $\tau = 0.5$, $\varepsilon = 0$ (linear in the delay). H_1 holds, H_2 a provable negative. Oracle: delayed-LQR (exact).
- **hopfield_nonlinear**. The same network with the saturating activation $\varphi(\xi) = \tanh(\kappa\xi)/\kappa$ on the delayed coupling, $\varphi'(0) = 1$ so the linearisation $A = -J$, $A_1 = W$ is unchanged: H_1 is held fixed while the value becomes a nonlinear functional of the history. The delay τ is the operative parameter for H_2 (it sets the window length). Oracle: Pontryagin.
- **hopfield_duffing**. The non-saturating hardening variant $\varphi(\xi) = \xi + \kappa\xi^3$ on the coupling, with an instantaneous cubic confinement $-d\,x^3$ to bound the plant; φ' *grows* with amplitude, so history sensitivity is preserved while the value is nonlinear. (κ, d) swept. Oracle: Pontryagin.

- **platoon**. Five vehicles in connected cruise control (optimal-velocity model), linearised at uniform flow; state $\mathbb{R}^{10}$, every feedback term delayed by $\tau = 0.2$. A high-*state-dimension* example. Oracle: delayed-LQR (linearised).
- **mg_limit_cycle**, **mg_chaotic**. The scalar Mackey–Glass oscillator $\dot{x} = -\mu x + p\,x(t-\tau)/(1 + x(t-\tau)^{10}) + u$, regulating to $x^* = 1$, at $\tau = 6$ (limit cycle) and $\tau = 17$ (chaos). Long-*delay* examples. Oracle: Pontryagin.

Table 1: Example parameters and feature dimensions. n is the state dimension, τ the delay, Δt the control step, L the window length; $\dim\phi$ from the measured `summary.json`. The raw-history dimension grows as $O((Ln)^2)$; the signature dimension is fixed in τ (it grows instead with n). "Oracle": R = delayed-LQR Riccati, P = Pontryagin boundary-value problem.

example	n	τ	Δt	L	$\dim\phi_M$	$\dim\phi_H$	$\dim\phi_S$	oracle
markovian	2	—	0.10	5	5	65	30	R
linear_dde	1	1.0	0.20	6	2	27	12	R
hopfield_linear	2	0.5	0.10	6	5	90	30	R
hopfield_nonlinear	2	0.5–6	0.10	6–61	5	90–7625	30	P
hopfield_duffing	2	0.5	0.10	6	5	90	30	P
platoon	10	0.2	0.05	5	65	1325	462	R
mg_limit_cycle	1	6.0	0.25	25	2	350	12	P
mg_chaotic	1	17.0	0.25	69	2	2484	12	P

Table 1 already contains the two structural facts the report turns on. The raw-history dimension $\dim\phi_H$ explodes with the *delay* (Hopfield $90 \to 7625$ as $L : 6 \to 61$), the engine of the H_2 illusion of Section 5. The signature dimension $\dim\phi_S$ is fixed in the delay but explodes with the *state dimension* ($12 \to 462$ from $n : 1 \to 10$), the engine of its failure on the platoon. Neither growth is a representational gain; Section 6 shows both are over-parameterisations of the same low-dimensional object.

3 The comprehensive learned benchmark

The primary benchmark is the comprehensive learned study (`main_unified`, June 10–11): each example is trained end-to-end (the value-gradient agent, semi-gradient temporal difference) across a sweep of representation capacity *— markovian and raw-history at polynomial degree* 1–3*, signature at depth* 2–4 *— over five seeds, and scored by the normalised sub-optimality* $\rho := (J - J^*)/|J^*|$ *against the per-example delayed-LQR oracle cost* J^* *(raw cost* J *for the nonlinear examples, where no closed-form* J^* *exists). This is a broader instrument than the controlled oracle-half protocol (Section 4): it sweeps capacity, not a fixed triple, and it measures the* learned *agent, so its verdicts carry the conditioning and optimisation behaviour of a real training run. The controlled protocol then resolves the ambiguities this benchmark exposes.*

Empirical finding 3.1 (H_1 holds strongly on the genuinely history-dependent examples)**.** On the examples where the delay couples the dynamics the learned markovian critic is one to three orders of magnitude worse than the best raw-history critic (Table 2): delayed_velocity 55.9 versus 0.142, mg_limit_cycle 30.3 versus 0.281, mg_chaotic 43.8 versus 0.297. History is decisive, measured over five seeds.

Remark 3.2 (Dadebo is near-Markovian and is demoted)**.** The originally-intended nonlinear-delayed example, the Dadebo continuous-stirred-tank reactor, is *measured* to be structurally near-Markovian: its linearised history-kernel ratio $\|K_{\text{hist}}\| / \|K_{\text{now}}\| = 0.009$ and the markovian-versus-history cost gap is +1% at the paper delays. Consistently, in Table 2 the markovian critic (0.027) is the *best* representation on this example, below every history and signature variant: it does

Table 2: Comprehensive learned benchmark (`main_unified`): per-example, per-capacity mean metric over five seeds. Metric S = normalised sub-optimality ρ (linear examples, lower better, 0 = oracle); C = raw closed-loop cost (nonlinear examples). Within each example the ordering is invariant to the metric (shared per-example baseline). Bold marks the best raw-history (versus markovian, H_1) and the best signature (versus best raw-history, H_2). "raw$_d$" is degree-d raw-history, "sig$_\ell$" depth-ℓ signature; a dash is a capacity not run.

example	m	mark	raw$_1$	raw$_2$	sig$_2$	sig$_3$	sig$_4$	H_1	H_2
delayed_velocity	S	55.9	1.32	**0.142**	**0.051**	0.109	–	**holds**	h.*
delayed_velocity_v2	S	0.572	1.28	**0.500**	**0.332**	0.330	–	**holds**	h.*
mg_limit_cycle	C	30.3	**0.281**	15.3	0.068	0.057	**0.049**	**holds**	h.*
mg_chaotic	C	43.8	1.11	**0.297**	**0.088**	0.106	0.114	**holds**	h.*
delayed_oscillator (high gap)	S	0.380	0.661	0.413	0.294	**0.158**	–	fails‡	h.*
linear_cell	S	**0.127**	0.711	0.322	**0.284**	3.86	–	fails‡	h.*
dadebo_cstr (demoted)	C	**0.027**	0.071	0.041	0.038	0.035	0.038	fails§	—
strongly_delayed	S	all representations diverge ($\rho \sim 10^9$)						—	—

*The learned-pipeline H_2 "hold" (signature lowest) on the long-delay examples is the under-sampling illusion overturned under control in Section 5. ‡On the ill-conditioned examples the learned raw-history is *under-converged*, not out-represented (see Finding 3.3). §Dadebo is near-Markovian (the markovian critic is the best representation); it is demoted to a stress example (Remark 3.2).

not exercise H_1. It is therefore demoted to a stress/robustness example, and the delay-coupled Hopfield network (Section 5) replaces it as the nonlinear-delayed primary example.

Empirical finding 3.3 (On ill-conditioned examples the learned H_1 failure is under-convergence)**.** On the high-gap oscillator and the long-window linear example the learned pipeline reports H_1 *failing* — markovian at or below raw-history (0.380 vs 0.413; 0.127 vs 0.322). This is an optimisation artefact, not a representational fact, and three independent measurements say so. First, the oracle-half plain-least-squares fit (Section 4) attains $R^2 \to 1$ for raw-history on these examples — it *spans* the value — and recovers H_1. Second, the dedicated budget sweep (Appendix C) shows the learned raw-history cost still decreasing with the training budget ($2.60 \to 2.32$ over a $9\times$ budget increase) while the markovian cost is flat (3.22): raw-history is budget-limited, markovian is representation-limited. Third, the feature-Gram condition number of raw-history exceeds the markovian one by orders of magnitude on these examples, so semi-gradient temporal difference needs proportionally more updates in the ill-conditioned directions. The H_1 "failure" is the conditioning mechanism of Section 6 acting on the learning dynamics.

Empirical finding 3.4 (The capacity sweep exposes over-parameterisation directly)**.** Sweeping capacity, absent from a fixed triple, shows the over-parameterisation of Section 6 as a non-monotonicity in degree and depth (Table 2). Degree-two raw-history *diverges* where degree-one survives on mg_limit_cycle (15.3 at $d = 2$, dimension 464, versus 0.281 at $d = 1$, dimension 29): the higher-dimensional map is under-sampled. Signature depth past the optimum degrades sharply (linear_cell: 0.284 at depth 2, 3.86 at depth 3): the approximation gain is overtaken by the conditioning loss. The interior capacity optimum is the bias/conditioning trade-off of the companion note's decomposition.

4 H_1 under the controlled protocol

The controlled cross-check is the canonical oracle-half run: five examples, five seeds each, on + off-sheet data, plain least squares to the oracle value (`20260616_161650_on_off_sheet_array`). Because the estimator solves the least-squares problem exactly, it removes the under-convergence of Finding 3.3 and isolates the representational question. The Markovian example is the negative control; the four delay examples are the test. The learning-half confirmation is in Appendix B.

4.1 The negative control

Empirical finding 4.1 (The Markovian example falsifies a spurious H_1)**.** On the Markovian double integrator the three representations are *measured to coincide*, $J = 3.1151 \pm 0.0000$ for markovian, raw-history and signature alike (five seeds; the off-sheet labels pin the history coefficients of ϕ_H to zero because V^* carries no history dependence). H_1 therefore fails here, as it must: history that is irrelevant is correctly *not* rewarded. The benchmark is thus not biased toward H_1.

Remark 4.2 (Off-sheet data is what makes the negative control exact)**.** On on-sheet-only data the same example gives raw-history $J = 3.1939 \pm 0.0385$ (a 2.5% degradation, gradient cosine 0.988): without off-manifold labels the redundant history coefficients are not fully pinned. The exactness of Finding 4.1 is a consequence of the off-sheet design of Definition 1.5, foreshadowing Section 6.

4.2 The delay examples

Table 3: Canonical H_1/H_2 table: closed-loop cost J, value-fit R^2 and gradient cosine η, mean $\pm$ half-range over five seeds (oracle half, on+off-sheet, plain least squares). J^0 is the no-control cost, J^* the oracle-feedback cost. Lower J is better. Bold marks the lower of raw-history vs markovian (H_1) and signature vs raw-history (H_2).

example ($J^0 \to J^*$)	rep	R^2	η	J	verdict
linear_dde ($3.89 \to 0.14$)	markovian	0.72 ± 0.11	0.633 ± 0.017	0.3645 ± 0.0056	
	raw-history	1.00 ± 0.00	0.998 ± 0.000	$\mathbf{0.1518 \pm 0.0000}$	H_1 **holds**
	signature	0.97 ± 0.02	0.957 ± 0.020	0.1840 ± 0.0159	H_2 fails
platoon ($0.075 \to 0.028$)	markovian	0.991 ± 0.001	0.434 ± 0.081	0.0842 ± 0.0232	
	raw-history	1.000 ± 0.000	0.988 ± 0.006	$\mathbf{0.0289 \pm 0.0003}$	H_1 **holds**
	signature	0.999 ± 0.000	0.318 ± 0.161	0.2272 ± 0.1216	H_2 fails
mg_limit_cycle ($0.52 \to 0.010$)	markovian	0.14 ± 0.04	0.051 ± 0.000	5.5965 ± 0.7720	
	raw-history	0.92 ± 0.02	0.377 ± 0.046	0.0359 ± 0.0111	H_1 **holds**
	signature	1.00 ± 0.00	0.938 ± 0.006	$\mathbf{0.0128 \pm 0.0002}$	H_2 **holds**†
mg_chaotic ($3.64 \to 0.013$)	markovian	0.09 ± 0.04	0.351 ± 0.001	5.34 ± 0.74	
	raw-history	0.95 ± 0.01	0.154 ± 0.134	1541 ± 949	H_1 fails‡
	signature	0.99 ± 0.00	0.838 ± 0.023	$\mathbf{0.0180 \pm 0.0008}$	H_2 **holds**†
markovian ($423 \to 3.12$)	markovian	1.00 ± 0.00	1.000 ± 0.000	3.1151 ± 0.0000	
	raw-history	1.00 ± 0.00	1.000 ± 0.000	3.1151 ± 0.0000	H_1 fails
	signature	1.00 ± 0.00	1.000 ± 0.000	3.1151 ± 0.0000	H_2 fails

†The signature's lower cost on the long-delay Mackey–Glass examples is the under-sampling artefact overturned in Section 5. ‡On mg_chaotic raw-history *diverges* ($J \sim 10^3$, all five seeds) for the same reason — its dimension 2484 is under-sampled at this sample size — so it falls below even the markovian critic; this is not evidence against history.

Empirical finding 4.3 (H_1 holds wherever the delay couples the dynamics)**.** On every example where the delay genuinely enters the dynamics, the history window lowers the closed-loop cost relative to the state-only critic, by a margin far exceeding the seed half-range (Table 3): linear_dde 0.152 versus 0.365 (−58%), platoon 0.029 versus 0.084 (−66%), mg_limit_cycle 0.036 versus 5.60 (two orders of magnitude). The markovian critic's deficit is visible already in the gradient cosine — $\eta = 0.05$ on mg_limit_cycle, 0.43 on platoon — the optimal control is simply not a function of the current state. H_1 is a robust, measured, seed-stable positive.

Remark 4.4 (H_1 is about the dynamics, not the representation)**.** On mg_chaotic the history critic *diverges* ($J \sim 10^3$), so the raw H_1 comparison inverts (Table 3, ‡). This is not a failure of history but a conditioning failure of the particular history *representation* at $\tau = 17$ (window 69, dimension 2484): the markovian critic, with no history to misfit, is merely the less-broken of two inadequate fits. The honest reading is that history *matters* (the markovian critic's $R^2 = 0.09$ shows it cannot represent the value at all) but that the degree-two history map is under-sampled here — exactly the phenomenon Section 5 controls for.

4.3 Off-manifold design as a precondition

Empirical finding 4.5 (On-sheet-only data does not identify the gradient)**.** Repeating the canonical run on *on-sheet-only* data (`20260616_161649_on_sheet_array`) leaves the low-dimensional examples intact but makes the high-dimensional fits diverge: raw-history rises to $J = 117.6$ (mg_limit_cycle), 784.5 (mg_chaotic), and the markovian and signature critics to $J = 235.9 \pm 90.1$ and 88.7 ± 75.6 on the platoon, against their on + off-sheet values of order 10^{-2}. The value fit can be near-perfect on-trajectory ($R^2 > 0.99$) while the deployed control is unbounded, because the data do not constrain the gradient *transverse* to the trajectory. Off-sheet relabelling (Definition 1.5) is therefore a precondition for any of the representations, and the rest of the report uses it throughout. This is the empirical face of Proposition 6.5.

5 H_2: does the signature structure help?

This is the overturned principal result. The measured empirical findings (the apparent signature advantage; its collapse under the controls) are kept strictly separate from the corrected mechanism (Section 6), so that refuting the latter would not cost the former. The controls below are all on `hopfield_nonlinear` *(the example whose delay τ is a free parameter) and the Mackey–Glass examples (a different plant family, to test generality).*

5.1 Formalising H_2

The comparison $J[\text{signature}] < J[\text{raw-history}]$ is, as stated, ill-posed: it depends on the data budget and the sampling law, which do not appear in it. Making them explicit separates two distinct questions. Fix a feature map $\phi : \mathcal{X} \to \mathbb{R}^{\dim \phi}$ on the space of history sequences $\mathcal{X} = (\mathbb{R}^n)^L$, and let

$$\mathcal{H}_\phi = \left\{ \underline{x} \mapsto \langle \theta,\, \phi(\underline{x}) \rangle : \theta \in \mathbb{R}^{\dim \phi} \right\}$$

be the induced linear value class. A critic V is deployed through Doya's law (3), $u_V = \frac{1}{2} R^{-1} B^\top \partial_x V$, and $J(V)$ denotes the resulting closed-loop cost. Because the control is the *gradient* of V, the operative metric is $J(V)$ — not any value-approximation error — which is why a perfect value fit ($R^2 = 1$) is uninformative on its own (Proposition 6.5). An estimator $\mathcal{D} \mapsto \hat{\theta}_\phi(\mathcal{D})$ on a dataset of size n (Definition 1.6) gives the fitted critic $\hat{V}_\phi^{\mathcal{D}}$. The two representations are ϕ_H (degree-two raw-history, $\dim \phi_H = O(\tau^2)$) and ϕ_S (depth-two signature, $\dim \phi_S$ fixed in τ). The hypothesis then admits two well-posed forms.

Population (representational) — does the signature structure enlarge the attainable control:

$$H_2^{\text{pop}} : \qquad \inf_{V \in \mathcal{H}_{\phi_S}} J(V) \;<\; \inf_{V \in \mathcal{H}_{\phi_H}} J(V). \tag{4}$$

Finite-sample (statistical), indexed by the budget n and the sampling law $\mathcal{P}$ (on/off-sheet, Definition 1.5):

$$H_2^{(n)} : \qquad \mathbb{E}_{\mathcal{D} \sim \mathcal{P}^{\otimes n}}\big[J(\hat{V}_{\phi_S}^{\mathcal{D}})\big] \;<\; \mathbb{E}_{\mathcal{D} \sim \mathcal{P}^{\otimes n}}\big[J(\hat{V}_{\phi_H}^{\mathcal{D}})\big], \tag{5}$$

well-defined only when $\mathcal{P}$ carries off-manifold support — otherwise $\ker G_n \neq \{0\}$ and both sides diverge (Proposition 6.5, Finding 4.5). For each class the excess cost decomposes as a bias/variance pair,

$$\mathbb{E}\big[J(\hat{V}_\phi)\big] - J^* = \underbrace{\Big(\inf_{V\in\mathcal{H}_\phi} J(V) - J^*\Big)}_{\text{approximation}} + \underbrace{\Big(\mathbb{E}\big[J(\hat{V}_\phi)\big] - \inf_{V\in\mathcal{H}_\phi} J(V)\Big)}_{\text{estimation (conditioning)}}. \tag{6}$$

The remainder of this section establishes, by measurement, that the *approximation* terms of ϕ_H and ϕ_S are comparable (both classes contain a near-optimal controller; degree-two raw-history spans V^* on the linear examples, $R^2 \to 1$), so H_2^{pop} does *not* hold; whereas the *estimation* terms differ and scale as $\dim\phi/n$, so $H_2^{(n)}$ holds only in the under-sampled regime $n \lesssim c\,\dim\phi_H$ and reverses for $n \gtrsim 2\,\dim\phi_H$ (the off/dim $\gtrsim 2$ transition, Table 5, Observation 6.9). The naive inequality, when it holds, therefore measures the estimation term at fixed budget — sample efficiency ($\dim\phi_S$ fixed versus $\dim\phi_H = O(\tau^2)$) — not expressivity; to "collapse" it under the controls is precisely to raise n past that threshold.

5.2 The apparent advantage

Table 4: Hopfield τ-sweep at fixed sample size $n_{\mathrm{train}} = 4130$ (single seed, `h1h2_tau_sweep`). As τ grows the window L and hence $\dim\phi_H$ grow as $O(\tau^2)$, so $n_{\mathrm{train}}/\dim\phi_H$ falls; below $\sim$5 the history fit loses the gradient (η collapses) and J *diverges*, leaving the fixed-dimension signature as the apparent winner.

τ	L	$\dim\phi_H$	$\frac{n}{\dim\phi_H}$	η_H	J_H (raw-hist)	η_S	J_S (signature)
0.5	6	90	45.9	1.000	**0.090**	0.960	0.098
1.0	11	275	15.0	0.997	**0.130**	0.921	0.152
2.0	21	945	4.4	0.166	544.6	0.812	**0.264**
3.0	31	2015	2.0	0.120	383.5	0.937	**0.314**

Empirical finding 5.1 (The signature has the lower cost only where the history fit is under-sampled)**.** In the τ-sweep at fixed sample size (Table 4) the history window has the *lower* cost at short delay ($\tau = 0.5, 1$: $J_H = 0.090, 0.130$ versus $J_S = 0.098, 0.152$), and the signature only overtakes it at $\tau = 2, 3$. But the overtaking coincides exactly with $n_{\mathrm{train}}/\dim\phi_H$ falling through $\sim$5 to $\sim$2: at $\tau = 2, 3$ the history *diverges* ($J_H = 544, 383$; $\eta_H = 0.17, 0.12$), it is not out-performed. The apparent H_2 advantage is a statement about sample size, not representation — a hypothesis to be controlled, not a result.

5.3 Three controls collapse the advantage

Empirical finding 5.2 (Every control overturns the apparent advantage)**.** Table 5 reports three independent controls. (a) Quadrupling the data at $\tau = 2$ restores the history window from $J = 544.6$ to 0.240 and its gradient cosine from 0.17 to 0.98, level with the signature's 0.229. (b) At fixed under-sampled data, Tikhonov conditioning alone restores it ($J : 544.6 \to 0.238$ at $\lambda = 10^{-2}$) — and *degrades* the signature ($J : 0.264 \to 6.887$), since the same ridge over-damps the signature's few informative directions. The cure is conditioning, not capacity. (c) At $\tau = 3$, scaling the off-sheet count until off/dim ≥ 2 makes the history window converge and *beat* the signature: $J_H = 0.289$ versus $J_S = 0.323$, with $\eta_H = 1.000$. Under any of the three controls the H_2 ordering of Table 4 reverses.

Empirical finding 5.3 (On/off-sheet *balance*, not raw count, is the lever)**.** The controls isolate a sharper sub-finding. At $\tau = 3$, adding on-sheet data makes things *worse*: with $n_{\mathrm{off}} = 2000$ fixed, raising the on-sheet count from $n_{\mathrm{ic}} = 30$ to 240 moves the history window from $J = 0.579$ (converging) to $J = 1495$ (diverged), at $n/\dim\phi_H = 15.4$ — a *larger* sample (`h1h2_tau3_control`). The on-sheet windows are low-rank (Section 6) and, in excess, they *dominate* the

Table 5: Three controls on `hopfield_nonlinear` at $\tau = 2$ and $\tau = 3$ (single seed). Each restores the history window from divergence to a cost at or below the signature's. (a) quadrupled data; (b) Tikhonov conditioning at fixed (under-sampled) data; (c) off-sheet rebalancing at fixed on-sheet count.

control	setting	$n/\dim\phi_H$	η_H	J_H	J_S
(a) more data — Hopfield $\tau = 2$, $\dim\phi_H = 945$ *(`h1h2_tau2_control`)*					
baseline (under-sampled)	$n_{\mathrm{ic}}30$, $n_{\mathrm{off}}500$	4.4	0.166	544.6	0.264
$\times 4$ data	$n_{\mathrm{ic}}120$, $n_{\mathrm{off}}2000$	17.5	0.984	**0.240**	0.229
(b) Tikhonov conditioning — Hopfield $\tau = 2$, *under-sampled data (`h1h2_tau2_ridge`)*					
$\lambda = 0$	(baseline)	4.4	0.166	544.6	0.264
$\lambda = 10^{-2}$	ridge	4.4	0.981	**0.238**	6.887
$\lambda = 10^{-1}$	ridge	4.4	0.977	**0.261**	35.92
(c) off-sheet rebalancing — Hopfield $\tau = 3$, $\dim\phi_H = 2015$ *(`h1h2_tau3_offsheet`)*					
$n_{\mathrm{off}}2000$ (off/dim 1)	$n_{\mathrm{ic}}30$	2.8	0.916	0.579	0.318
$n_{\mathrm{off}}4000$ (off/dim 2)	$n_{\mathrm{ic}}30$	3.8	0.998	**0.289**	0.333
$n_{\mathrm{off}}8000$ (off/dim 4)	$n_{\mathrm{ic}}30$	5.8	1.000	**0.289**	0.323

least-squares normal equations and suppress the off-sheet gradient constraint. What conditions the fit is the *balance* off/dim, not n_{train}: the same $n_{\mathrm{off}} = 2000$ gives divergence at $n_{\mathrm{ic}} = 240$ and convergence at $n_{\mathrm{ic}} = 30$.

5.4 Generality: a different nonlinearity and a different plant

Table 6: Left: Duffing κ-sweep (`h1h2_duffing_kappa_sweep`), a non-saturating hardening nonlinearity at short delay (window 6, $\dim\phi_H = 90$, well sampled at $n/\dim\phi_H = 45.9$). Right: Mackey–Glass off-sheet scaling (`h1h2_mg_offsheet`, single seed), off/dim $\in \{1, 2, 4\}$. Lower J in bold.

κ	J_M	J_H	J_S
0.5	0.170	**0.104**	0.120
1.0	0.223	**0.121**	0.133
2.0	0.279	**0.167**	0.178
3.0	0.345	**0.206**	0.235

Duffing: H_1 holds, H_2 fails at every κ.

example	off/dim	η_H	J_H	J_S
	1	0.168	0.240	**0.0128**
limit cycle	2	0.589	0.0171	**0.0125**
	4	0.810	**0.0108**	0.0126
	1	0.305	0.193	**0.0179**
chaotic	2	0.869	**0.0143**	0.0184
	4	0.898	**0.0140**	0.0180

Empirical finding 5.4 (H_2 fails on a well-sampled nonlinear example)**.** On the Duffing example, where the delay is short ($\dim\phi_H = 90$) so the history window is *well* sampled at every κ, the history window has the lower cost at every nonlinearity strength (Table 6, left: $J_H < J_S$ throughout, $\eta_H \geq 0.97$ versus $\eta_S \leq 0.91$). A genuinely nonlinear value functional is therefore *not* sufficient for H_2: when the history fit is not under-sampled, the degree-two polynomial of the window already captures the nonlinearity the signature was meant to supply.

Empirical finding 5.5 (The under-sampling explanation generalises to Mackey–Glass)**.** On the Mackey–Glass limit cycle, scaling the off-sheet data reproduces the Hopfield pattern in a different plant family (Table 6, right): the history window moves from $J = 0.240$ (off/dim 1, $\eta_H = 0.17$, under-conditioned) through 0.017 (off/dim 2) to **0.0108** (off/dim 4), *beating* the signature's 0.0126. The canonical H_2-holds verdict on this example (Table 3, †) is thus an off-sheet-budget artefact: at off/dim = 4 it reverses. The chaotic example (window 69, $\dim\phi_H = 2484$, the hardest plant) overturns *sooner*: under-conditioned at off/dim 1 ($J_H = 0.193$ versus the signature's 0.0179), the history window already *beats* the signature at off/dim 2 ($J_H = 0.0143$ versus 0.0184, $\eta_H = 0.87$) and *plateaus* there ($J_H = 0.0140$, $\eta_H = 0.90$ at off/dim $= 4$; each such point requires $\sim 10^4$ Pontryagin solves). The reversal thus holds on both Mackey–Glass examples and both Hopfield delays: the signature's long-delay advantage is under-sampling of the history window, removed by off-sheet data balanced to its dimension, with no example measured as an exception.

Observation 5.6 (Verdict on H_2)**.** Across every example measured, the signature's advantage over the history window appears only where the history feature dimension is under-sampled, and is removed by data, by conditioning, or by off-sheet rebalancing. On well-sampled examples (Duffing, short-delay Hopfield) and on the high-state-dimension example (platoon) the history window is at least as good, and the signature is markedly worse on the platoon. H_2 — that the signature *structure* improves on a polynomial of the history — is not supported. What remains is a *sample-efficiency* advantage, quantified next.

6 Mechanism: density is not realisability

The empirical findings above are explained by a single structural fact, proven and then measured. The proven part (a Veronese effective-rank bound, and the value/gradient gap) is the companion note `latex_documents/notes/2026_06_17_signature_density_vs_conditioning`*; its operative statements are reproduced and cited here, not re-proved (the density theorem is a classical result, attributed there to Arribas and to Hambly–Lyons), with the measured part kept in a separate* Empirical finding.

6.1 The proven part

Theorem 6.1 (Density of linear functionals of the signature; Arribas 2018)**.** *Let $\mathcal{X}$ be a set of continuous bounded-variation paths $[0,T] \to \mathbb{R}^{1+d}$, time-augmented (each carrying a monotone time coordinate), compact for the 1-variation topology, and containing no two distinct tree-like-equivalent paths. Then for every $F \in C(\mathcal{X};\mathbb{R})$ and every $\varepsilon > 0$ there exist a truncation level $L \in \mathbb{N}$ and a linear functional ℓ of the depth-L signature such that*

$$\sup_{X\in\mathcal{X}} \left| F(X) - \langle \ell, \mathbb{S}_{\le L}(X) \rangle \right| < \varepsilon.$$

Remark 6.2 (Existence, not recovery)**.** Theorem 6.1 is an existence statement in $\big(C(\mathcal{X}), \|\cdot\|_\infty\big)$: it asserts that an ε-approximant ℓ *exists*, but bounds neither its norm, nor the stability of recovering it from a finite sample, nor any derivative of F. The obstructions of this section — the effective-rank degeneracy (Proposition 6.3) and the value/gradient gap (Proposition 6.5) — all live in that gap.

Proposition 6.3 (Veronese effective-rank bound)**.** *Let the law of the history sequence be supported on a family $\{\underline{x}^{(z)} : z \in \mathbb{R}^k\}$ indexed by a k-dimensional parameter z (for a regulated linear closed loop, z is the low-dimensional initial condition and $\underline{x}^{(z)}$ is affine in z). If each depth-m signature coordinate $z \mapsto \mathbb{S}^m(\underline{x}^{(z)})$ is a polynomial of degree $\le m$ in z, then the covariance of the depth-L signature features has rank bounded by the dimension of the Veronese embedding,*

$$\operatorname{rank}\operatorname{Cov}\left(\mathbb{S}_{\le L}(\underline{x})\right) \;\le\; \binom{k+L}{L},$$

independently of the ambient feature dimension $N(d,L) = \sum_{m\le L} d^m$. *The identical bound, with $L = 2$, governs the degree-two raw-history covariance. (Companion note, Proposition 3.2 and Corollary 3.3.)*

Lemma 6.4 (Differentiation is unbounded)**.** *There is a sequence $g_n \in C^1$ with $\|g_n\|_\infty \to 0$ and $\|g_n'\|_{L^2} \to \infty$ (e.g. $g_n(x) = n^{-1}\sin(n^2 x)$). A uniformly small error in the value is compatible with an arbitrarily large error in its gradient.*

Proposition 6.5 (Gradient indeterminacy on a degenerate design)**.** *Let $\hat{\ell}$ minimise the empirical value risk $\sum_i(\langle \ell, \Phi_i\rangle - V_i^*)^2$ and suppose the empirical Gram has a non-trivial kernel,* $\ker G_n \neq \{0\}$ *(the generic situation by Proposition 6.3, since the effective rank is $\ll N$). Then $\hat{\ell}+h$ is an equally good minimiser for every $h \in \ker G_n$, with identical fitted values on the sample; if some such h has non-zero gradient at a deployment point off the data manifold, the deployed control gradient there is* not determined by the data. *A perfect value fit ($R^2 = 1$) is compatible with an arbitrary control. (Companion note, Proposition 6.2.)*

Proposition 6.5 is the exact form of Finding 4.5: on-sheet-only data leaves $\ker G_n$ large, the value fits perfectly, and the off-manifold gradient — hence the control — is free, so J diverges. Off-sheet relabelling (Definition 1.5) shrinks $\ker G_n$ and restores identifiability. The natural objection is that exploration should do the same; the next two statements show why, on a regulated plant, it does not — and isolate the one thing off-sheet design does that no exploration can.

Proposition 6.6 (Exploration amplitude-invariance; companion note)**.** *For a Hurwitz closed loop with generator A, the stationary state covariance under exploration of noise covariance $\Sigma \succeq 0$ is the solution $P(\Sigma)$ of $AP + PA^\top + \Sigma = 0$. For a fixed noise shape Σ_0, the effective rank* $\operatorname{erank} P(\sigma^2\Sigma_0)$ *and the condition number* $\operatorname{cond} P(\sigma^2\Sigma_0)$ *are independent of the exploration amplitude σ. (Companion note, Proposition 4.1; the map $\Sigma \mapsto P(\Sigma)$ is linear, so $P(\sigma^2\Sigma_0) = \sigma^2 P(\Sigma_0)$, and* $\operatorname{erank}, \operatorname{cond}$ *are degree-zero homogeneous.)*

Observation 6.7 (Off-sheet design versus exploration)**.** Off-sheet perturbation and exploration pursue the same end — populating the directions transverse to the optimal trajectory, on which the value gradient is otherwise unconstrained (Proposition 6.5) — yet they are not interchangeable, and the distinction is structural.

1. *Amplitude does not condition.* By Proposition 6.6 the stationary distribution reached by isotropic exploration ($\Sigma_0 = \mathrm{Id}$) has an effective rank set by the closed-loop spectrum, not by the amplitude σ; with the Veronese bound (Proposition 6.3) the induced feature Gram inherits the same bounded effective rank as the on-sheet attractor. Increasing isotropic exploration therefore cannot lift $\ker G_n$ — the refuted remedy "add more noise".

2. *Shape can; exploration is not futile.* The invariance is in the *amplitude* at a fixed noise shape; an anisotropic exploration $\Sigma_0 \neq \mathrm{Id}$, chosen to excite the weak directions, yields a different $P(\Sigma_0)$ and *can* change the effective rank. What fails is the naive isotropic-amplitude form, not directed exploration.

3. *Off-sheet decouples the query law from the flow.* The support of any exploration-driven occupancy measure lies in the reachable set $\mathcal{R}$ of the closed loop: every state it labels is one the dynamics can produce. Off-sheet relabelling instead evaluates V^* at query points $\underline{x}' = \underline{x} + \varepsilon\,\Xi$ (Definition 1.5) that need not lie in $\mathcal{R}$ — the oracle solves the control problem from $\underline{x}'$ as an initial condition, whatever its reachability. Off-sheet is thus state-space sampling that the stabilising loop does not contract, and it can populate directions transverse to $\mathcal{R}$ that no exploration supported on $\mathcal{R}$ can reach.

The operative lever is off-manifold *design*; its one component that no exploration can replicate — querying the value *off* the reachable set — is structural and requires an oracle or a model, not a larger exploration amplitude.

6.2 The measured part

Empirical finding 6.8 (The on-manifold effective rank is flat while the dimension explodes)**.** For the Hopfield example the on-sheet (controlled-trajectory) windows were generated as in the benchmark and the centred degree-two raw-history feature covariance formed; its effective rank $\operatorname{erank} = (\operatorname{tr} C)^2 / \|C\|_F^2$ (participation ratio) is reported against the nominal dimension in Table 7 and Figure 1. As the delay grows from $\tau = 1$ to $\tau = 6$ the dimension increases by a factor of 28 ($275 \to 7625$) while the effective rank rises only from 3.32 to 3.50 (a 5% change). The history feature Gram is thus over-parameterised by a factor dim / erank rising from ∼83 to ∼2180: it depends entirely on the off-sheet data for conditioning, which is why the balance of Finding 5.3, not the raw count, governs it. The signature's effective rank is likewise flat (near 2.6) against its fixed dimension 30 — the same low-dimensional object, embedded with fewer coordinates.

Table 7: On-manifold effective rank of the degree-two raw-history feature covariance, Hopfield example, 30 optimal-trajectory initial conditions (the benchmark on-sheet count), $n_{\text{win}} = 3630$ windows (`h1h2_effective_rank_vs_delay`). The dimension grows as $O(\tau^2)$ by a factor 28; the effective rank rises only from 3.32 to 3.50. The signature columns (fixed dimension 30) are shown for contrast.

τ	L	$\dim \phi_H$	erank C_H	dim / erank	erank C_S
1	11	275	3.32	83	2.55
2	21	945	3.31	285	2.60
3	31	2015	3.34	603	2.60
4	41	3485	3.39	1028	2.59
5	51	5355	3.44	1557	2.58
6	61	7625	3.50	2179	2.56

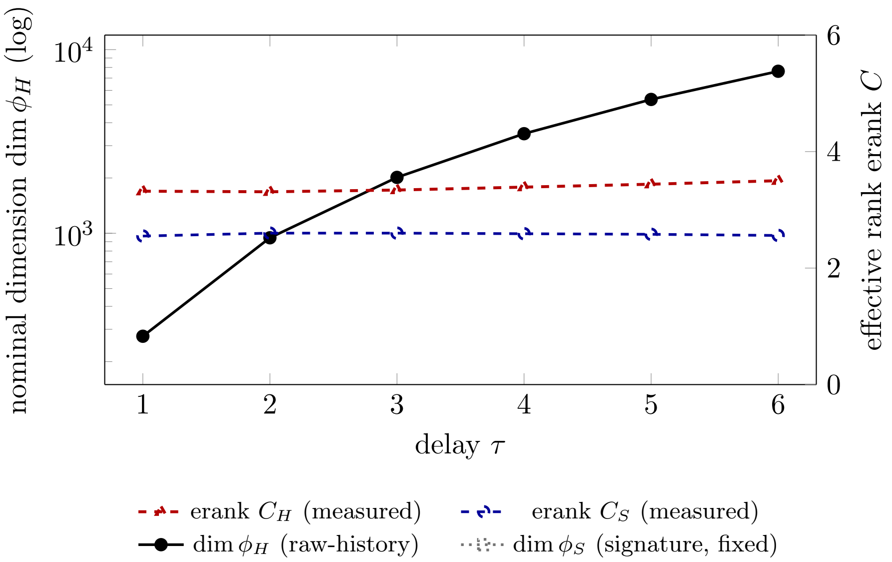

Figure 1: The mechanism in one figure. The raw-history nominal dimension (solid, left log axis) grows by 28× over $\tau \in \{1, \ldots, 6\}$; the on-manifold effective rank (dashed, right linear axis) rises only from 3.3 to 3.5. The controlled trajectory occupies a ∼3-dimensional set regardless of how many feature coordinates encode it, in accordance with the Veronese bound of Proposition 6.3. Solid = nominal dimension; dashed = measured effective rank; dotted = the signature's fixed dimension.

Observation 6.9 (The off-sheet budget scales with the feature dimension: the sample-efficiency law)**.** Gradient identifiability requires $\ker G_n = \{0\}$ (Proposition 6.5). The on-sheet Gram has rank equal to the effective rank $r^* = \operatorname{erank} C \approx 3.4$ (Finding 6.8, the Veronese bound of Proposition 6.3), so its co-rank is $\dim \phi - r^*$. Off-sheet windows supply the missing rows, each raising $\operatorname{rank} G_n$ by at most one; reaching full rank therefore needs

$$n_{\text{off}} \;\gtrsim\; \dim \phi - r^* \;=\; \dim \phi \,\big(1 + o(1)\big) \qquad (\dim \phi \to \infty).$$

The off-sheet perturbations are of small amplitude, hence non-generic, so each contributes less than a full unit of rank and the measured threshold is $n_{\text{off}} \gtrsim c \dim \phi$ with $c \approx 2$ (the off/dim ≈ 2 transition of Table 5), not $c = 1$. Since the signature dimension is fixed while the degree-two history dimension grows as $\dim \phi_H = O(\tau^2)$ (window length $L = O(\tau)$, $\dim \phi_H = O((Ln)^2)$),

$$n_{\text{off}}^{\text{sig}} = O(1), \qquad n_{\text{off}}^{\text{raw}} = O(\tau^2) \qquad (\tau \to \infty).$$

This is the quantitative content of the sample-efficiency advantage (Observation 5.6): the signature is not more expressive; the off-sheet data it requires for identifiability is independent of the delay, whereas the history window's grows quadratically. The statement is falsifiable — it predicts, for any τ, the off-sheet count at which the history fit recovers — and is corroborated by the off/dim $\gtrsim 2$ transitions measured on both the Hopfield and the Mackey–Glass examples.

Empirical finding 6.10 (On-sheet data identifies only the low-dimensional critic; off-sheet data is a shared precondition)**.** The effective rank of the on-sheet (controlled-trajectory) data is ≈ 3.4 (Finding 6.8), so any representation whose feature dimension exceeds ≈ 3.4 is under-determined by on-sheet data alone: the value fits ($R^2 \to 1$) while the gradient transverse to the trajectory is unconstrained (Proposition 6.5). This is measured directly on the Hopfield example with on-sheet data only, over five seeds (Table 8), extending the canonical on-sheet-only comparison of Finding 4.5 to the primary nonlinear-delayed example and tying it to the rank. Only the dimension-5 markovian critic — whose dimension is close to the attractor rank — is identifiable and stabilises the plant ($J = 0.13$ at $\tau = 0.5$ and 0.64 at $\tau = 2$, against the oracle 0.09 and 0.21); both high-dimensional representations diverge ($J \sim 3$–4×10^2, all five seeds), exceeding even the no-control cost ($J^0 = 9.85$ and 33.5). The decisive case is $\tau = 0.5$, where raw-history is *over-sampled in count* ($n/\dim = 40$) yet still diverges: the obstruction is the rank-≈ 3.4 degeneracy of the data, not the sample size. Off-sheet (off-manifold) data is therefore a precondition *shared by both* high-dimensional representations, and this shared precondition — not any property of the signature — is the reason on-sheet-only fits fail.

Table 8: Hopfield on-sheet-only (no off-sheet data): closed-loop J, mean$\pm$half the 95% confidence interval over five seeds (`data/h1h2_hopfield_onsheet_5seed`). Only the dimension-5 markovian critic (dimension near the on-sheet effective rank ≈ 3.4) is identifiable; both high-dimensional representations diverge — the signature marginally worse than raw-history — above even the no-control cost J^0. At $\tau = 0.5$ raw-history has $n/\dim = 40$, so the divergence is the rank degeneracy, not a sample shortage.

τ ($J^0 \to J^*$)	rep	$\dim \phi$	$n/\dim$	J (mean $\pm$ ci)
	markovian	5	726	$\mathbf{0.131 \pm 0.012}$
0.5 ($9.85 \to 0.09$)	raw-history	90	40.3	318 ± 8
	signature	30	121	393 ± 10
	markovian	5	726	$\mathbf{0.642 \pm 0.18}$
2.0 ($33.5 \to 0.21$)	raw-history	945	3.8	371 ± 27
	signature	30	121	415 ± 11

Observation 6.11 (The shared precondition sets up the efficiency distinction, not an expressivity gap)**.** Finding 6.10 is the precondition common to both high-dimensional maps; they differ only in *how much* off-sheet data restores them. Given off-sheet data, the scaling sweeps (Table 5, Table 6) show the signature needs less: its fixed dimension (~ 30) is over-determined already at the smallest off-sheet budget, whereas raw-history's $O(\tau^2)$ dimension requires off/dim $\gtrsim 2$ against its own growing count. Once that adequate, balanced off-sheet data is supplied, raw-history converges and matches or beats the signature on every example (Findings 5.2 and 5.5). The signature's advantage is therefore sample efficiency and conditioning, not expressivity: both high-dimensional representations require off-sheet data; the signature simply requires less of it.

Empirical finding 6.12 (The signature is sample-efficient but not exempt from the off-sheet precondition)**.** Quantitatively the signature's frugality is its over-parameterisation ratio $\dim \phi_S / \operatorname{erank} C_S \approx 12$, *fixed* in the delay (Table 7: dimension 12–30 over an on-manifold rank ≈ 2.6), against raw-history's 83–2180; by the co-rank bound (Observation 6.9) its off-sheet budget for identifiability is $O(1)$, not $O(\tau^2)$. On the low-state-dimension examples this frugality is extreme: *with no off-sheet data at all* the signature already attains its full on + off-sheet cost on both Mackey–Glass examples ($J = 0.015$–0.019) and on linear_dde — exactly where raw-history *diverges* ($J = 118$–785), Table 9. But the signature is not exempt from the identifiability obstruction: its dimension (12–30) still exceeds the on-manifold rank (≈ 3.4), so $\ker G_n \neq \{0\}$ on-sheet-only, and on the high-state-dimension platoon it too *diverges* without off-sheet data ($J = 88.7$ versus 0.23 with it; the Hopfield examples behave likewise, Table 8). Whether the signature's undetermined off-manifold gradient happens to stabilise is the example-dependent accident of Proposition 6.5: the signature is more sample-efficient, not immune.

Table 9: On-sheet-only (no off-sheet data) versus on + off-sheet closed-loop cost J, per example (canonical five-seed oracle half). Without off-sheet data the signature attains its on + off cost on the low-state-dimension examples (linear_dde, both Mackey–Glass) — where raw-history *diverges* — but diverges itself on the platoon; the Hopfield examples behave like the platoon (Table 8).

example	signature on-sheet	signature on+off	raw-history on-sheet	raw-history on+off
linear_dde	0.166	0.184	0.152	0.152
markovian	3.116	3.115	3.194	3.115
mg_limit_cycle	**0.0154**	0.0128	117.6	0.036
mg_chaotic	**0.0190**	0.0180	784.5	1541
platoon	88.7	0.227	0.029	0.029

6.3 Synthesis

Observation 6.13 (The signature trades a delay blow-up for a dimension blow-up)**.** The two representations encode the *same* low-effective-rank object (Proposition 6.3, Finding 6.8) with differently growing numbers of coordinates. The raw-history dimension grows with the *delay* as $O((Ln)^2) = O(\tau^2)$ (Table 1, Hopfield $90 \to 7625$); the signature dimension grows with the *state dimension* as $O(d^L)$ (platoon $12 \to 462$). Neither growth adds representational power; each is an over-parameterisation that the off-sheet data must condition. Hence the measured pattern: the signature's fixed delay-dimension helps on long-delay, low-state-dimension examples *when data are scarce* (its sample-efficiency advantage), and its channel growth hurts on the high-state-dimension platoon (where $\eta_S = 0.32$ and $J_S = 0.227$ exceed even the markovian critic). Given off-sheet data balanced to the representation's dimension (off/dim $\gtrsim 2$), the degree-two history window matches or beats the signature on every example measured. The signature's genuine advantage is sample efficiency, not representational power.

7 Conclusion

The history of the state is a genuine and robust ingredient of optimal control for delayed plants: H_1 holds, measured and seed-stable, on every example where the delay couples the dynamics, and fails exactly on the Markovian negative control and on the near-Markovian Dadebo reactor (demoted on the measured history-kernel ratio). Where the comprehensive *learned* benchmark reports H_1 failing on an ill-conditioned example, the controlled plain-least-squares protocol shows the cause to be under-convergence of the high-dimensional history representation, not a representational deficit: the learned and controlled instruments agree once conditioning is accounted for. The path-signature *structure*, by contrast, confers no representational advantage over a plain degree-two polynomial of the history window: every apparent advantage is an artefact of the history map being under-sampled at long delay, and is removed by more data, by conditioning, or by off-sheet rebalancing. The mechanism is the effective-rank phenomenon — proven as a Veronese bound, measured as a flat effective rank against a 28× dimension growth — which makes both representations over-parameterisations of a ∼3-dimensional controlled-trajectory object. The signature's one real benefit is a fixed feature dimension, i.e. sample efficiency at long delay and low state dimension; that benefit reverses at high state dimension, where the signature's own channel growth makes it the ill-conditioned choice. For deployment, the operative levers are off-manifold (off-sheet) data design and conditioning, not a richer feature algebra.

A Visualisation: controlled trajectories and control signals

Each figure overlays, from the example's deployment initial condition, the closed-loop state $x_0(t)$ (top) and control $u_0(t)$ (bottom) of the three critics against the analytic oracle reference and the no-control response. The critics are fitted on exactly the benchmark data (same seed, same protocol); the local closed-loop cost reproduces the cluster **`summary.json`** *value (e.g. linear_ dde markovian/raw/signature* 0.366/0.152/0.201*, matching the canonical seed-0 fit to four decimals), so these are faithful regenerations. Convention:* solid *= learned critics,* dashed *= analytic oracle,* dotted *= no control.*

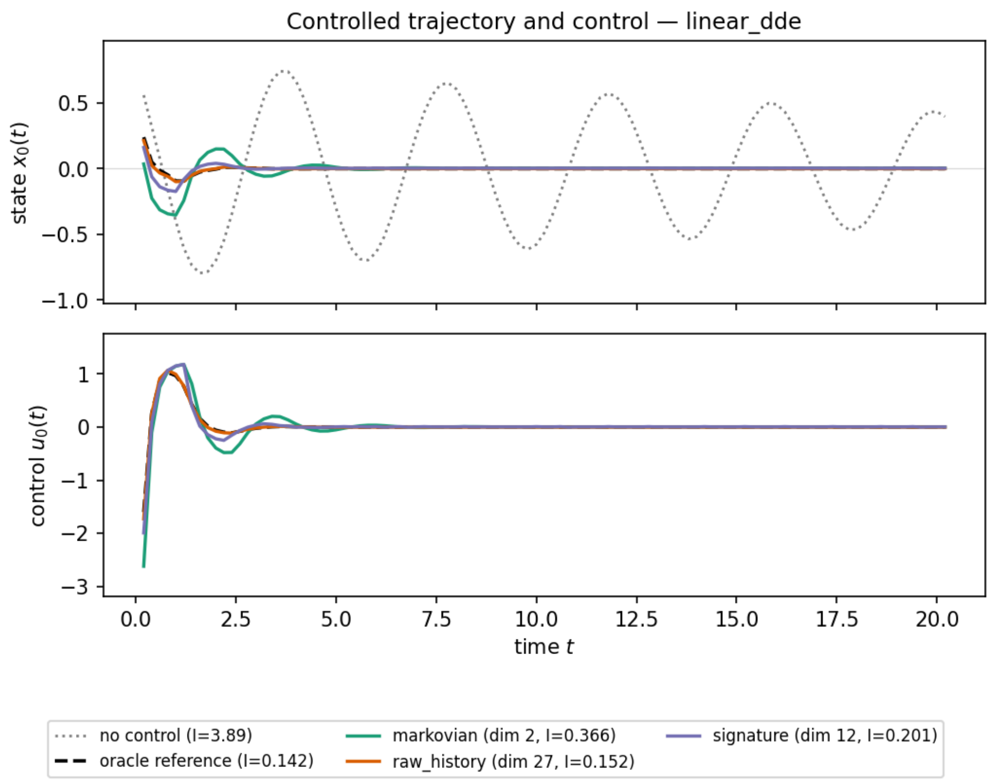

Figure 2: `linear_dde`. H_1 visible: the markovian critic (green) overshoots and settles slowly; the history window (orange) tracks the oracle and attains the lowest cost; the signature (purple) is marginally worse (H_2 fails). The no-control response (dotted) sustains an oscillation.

B Learning-half confirmation (LSPI)

The oracle half is the primary evidence; the damped-LSPI learning half confirms it under a temporal-difference estimator rather than a supervised one. These runs are single-seed (the full

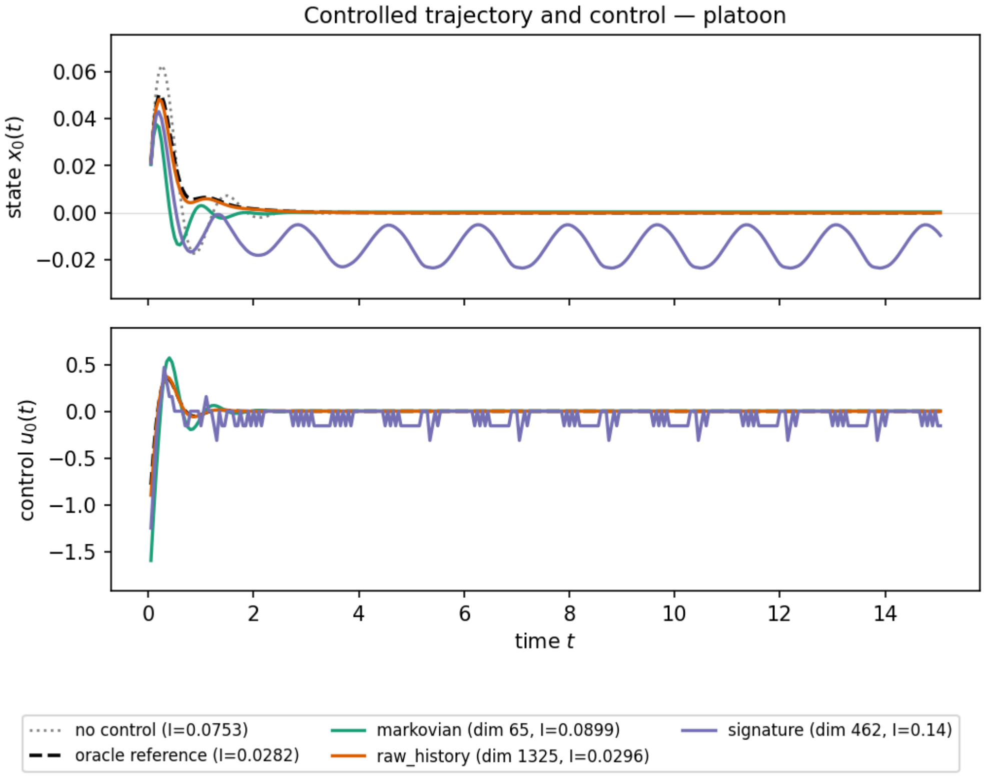

Figure 3: `platoon`. The decisive H_2 failure: the history window (orange, dimension 1325) tracks the oracle and settles; the signature (purple, dimension 462) *sustains an undamped oscillation* it never regulates, its control signal (bottom) saturating — the ill-conditioned gradient ($\eta_S = 0.32$, condition number $\sim 10^{14}$) producing a wrong feedback despite a near-perfect value fit ($R^2 = 0.999$).

learning-half matrix was cancelled on a storage quota), so they are confirmatory, not falsifiable. The $reg = 10^{-4}$ *conditioning fix of the parallel code is in effect.*

Table 10: Learning half (damped LSPI), closed-loop J, single seed (`h1h2_rl_lspi`). H_1 holds on the delay examples and fails on the Markovian negative control, mirroring Table 3.

example	J_M	J_H	J_S	H_1
markovian (neg. control)	3.782	3.687	3.849	fails
linear_dde	0.423	0.181	0.128	**holds**
hopfield_linear	0.134	0.108	0.111	**holds**
hopfield_duffing	0.147	0.118	0.115	**holds**

Observation B.1 (The learning half agrees on H_1; H_2 is inconclusive single-seed)**.** Table 10 reproduces the H_1 verdict: history lowers J on every delay example and not on the Markovian control. On H_2 the single-seed learning runs are mixed (signature marginally below history on linear_dde and hopfield_duffing, marginally above on hopfield_linear) and at the 3% margin are within run-to-run variation; no falsifiable H_2 claim is made from the learning half. The oracle half (Section 5), with five seeds and the controls, is the operative evidence.

C Robustness of the comprehensive benchmark

Three control studies on the high-gap oscillator support the readings of Section 3: an oracle ladder isolating the actor/critic contributions to the residual cost, a training-trick ablation, and a budget

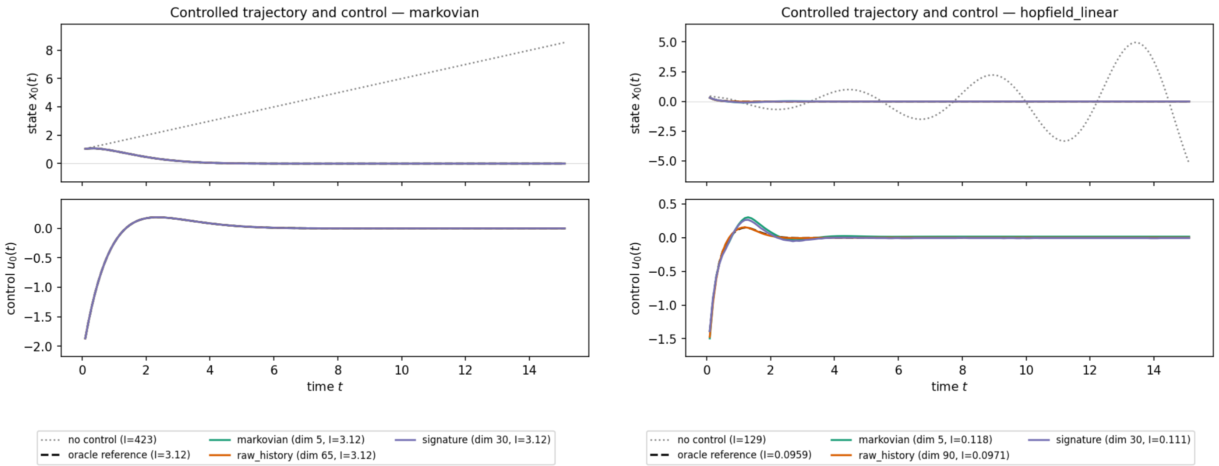

Figure 4: Left, `markovian` negative control: the three critics coincide (history irrelevant). Right, `hopfield_linear`: H_1 holds (history window below markovian), H_2 fails (signature above history window) — the provable-negative linear-in-delay arm.

sweep. All use the learned pipeline; costs are the best-iterate evaluation cost J (not the normalised ρ), so they are compared only within each study.

Table 11: Left: oracle ladder (`oracle_ladder_high_gap`, 3 seeds) — evaluation cost with the actor and/or critic replaced by the oracle. Right: budget sweep (`budget_sweep_high_gap`, 2 seeds) — evaluation cost versus training episodes.

configuration	J (best iterate)
full learned (actor + critic)	2.82 ± 0.08
oracle critic, learned actor	2.56 ± 0.03
oracle actor, learned critic	1.83 ± 0.00

Replacing the actor cuts the cost most ($2.82 \to 1.83$), the critic less ($2.82 \to 2.56$): the residual is dominated by policy (gradient) extraction, consistent with the value/gradient gap of Proposition 6.5.

budget	markovian$_2$	raw-history$_2$
800	3.22	2.60
2400	3.22	2.46
7200	3.22	2.32

Raw-history keeps converging with budget; markovian is flat (representation-limited). Evidence that the learned H_1 failure on this example is under-convergence (Finding 3.3).

Observation C.1 (The convergence is robust to the training tricks)**.** A five-way ablation of the stabilisers (`trick_ablation_convergence`, 3 seeds) leaves both the convergence speed and the final cost within a narrow band: episodes to the $\rho = 0.1$ threshold are 160–190, and the final ρ is 0.025 (minimal, no-target-net) to 0.036 (baseline, no-divergence-threshold, white-noise). Removing the target network or the divergence threshold does not break convergence; injecting white observation noise does not change the final cost beyond the seed band. The benchmark's verdicts are not artefacts of a particular training stabiliser.

D Reproduction

Every number and figure traces to a saved run. Scripts are in `run/study/`; oracle-half run directories are under `data/h1h2_evaluation/` (cluster `$WORK`) and the τ/control sweeps under the cluster `$SCRATCH` (`/lustre/fsn1/projects/rech/oym/ucd32aq`); they are consolidated locally under `data/h1h2_report_runs/`.

Scripts.

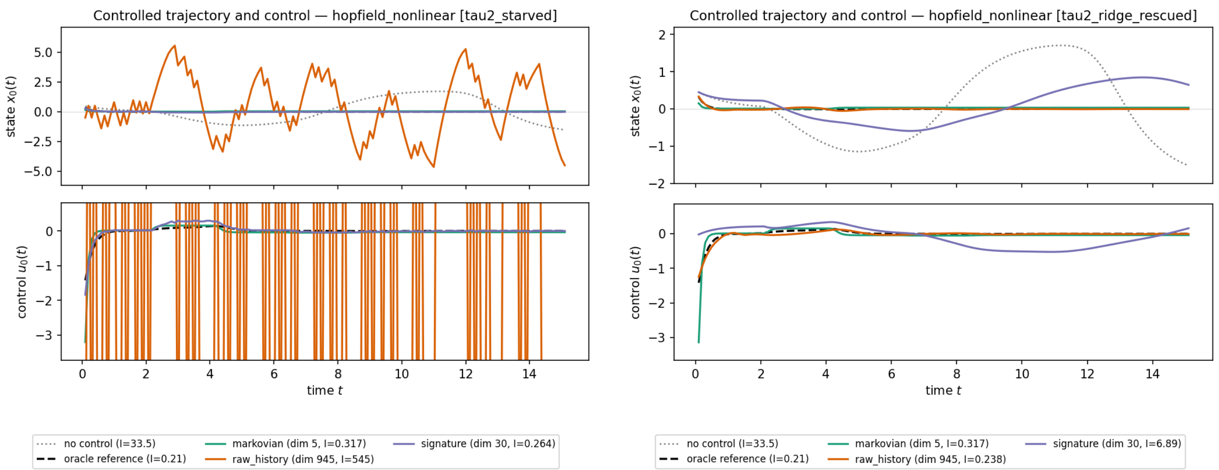

Figure 5: `hopfield_nonlinear` at $\tau = 2$: the H_2 illusion and its cure. Left, the under-sampled fit ($n/\dim\phi_H = 4.4$): the history window (orange, dimension 945) diverges into a saturating bang-bang control ($J = 545$) while the signature (purple) and oracle (dashed) track, so the signature appears preferable. Right, the same data with Tikhonov conditioning ($\lambda = 10^{-2}$): the history window is restored to $J = 0.238$ and tracks the oracle, while the identical ridge over-damps the signature ($J = 6.89$). The cure is conditioning, not capacity (Finding 5.2).

- `h1h2_evaluation.py` — oracle-half harness (examples, oracle, data protocol, plain least squares, three metrics). Tables 3, 4, 5, 6.
- `h1h2_rl_lspi.py` — learning-half (damped LSPI). Table 10.
- `h1h2_effective_rank_vs_delay.py` — on-manifold effective rank versus delay. Table 7, Figure 1.
- `h1h2_visualise_controlled_trajectories.py` — closed-loop trajectory/control figures (Appendix A); regenerates from saved `.npz`.
- Solvers: `src/solvers/continuous_oracle.py` (delayed-LQR), `{mackey_glass,delayed_hopfield}_pontryagin.py` (nonlinear oracle).

Run directories. All consolidated read-only under `data/h1h2_report_runs/` (index in `INDEX.md`); pulled from Jean Zay `$WORK` (learned/oracle) and `$SCRATCH` (sweeps).

- Comprehensive learned benchmark (Section 3, Table 2): `main_unified/` — per-example sub-studies (`*_study/summary.yaml`) sweeping capacity × 5 seeds. Robustness (Appendix C): `main_unified/{oracle_ladder_high_gap, trick_ablation_convergence, budget_sweep_high_gap_b*}`.
- Canonical H_1/H_2 (5 examples × 5 seeds): `20260616_161650_on_off_sheet_array` (Table 3); on-sheet-only twin `20260616_161649_on_sheet_array` (Finding 4.5).
- Hopfield: `h1h2_tau_sweep_20260617_165104` (Table 4); `h1h2_tau2_control_20260617_171721`, `h1h2_tau2_ridge_20260617_180843`, `h1h2_tau3_control_20260617_194242`, `h1h2_tau3_offsheet_20260617_204438` (Table 5, Finding 5.3).
- Duffing: `h1h2_duffing_kappa_sweep_20260617_155613`; Mackey–Glass off-sheet: `h1h2_mg_offsheet_20260630_164840` (Table 6).
- Effective rank: `data/h1h2_effective_rank_vs_delay/` (Table 7). Hopfield on-sheet-only (5 seeds): `data/h1h2_hopfield_onsheet_5seed/` (Table 8), produced by `h1h2_evaluation.py --data-mode on_sheet`.

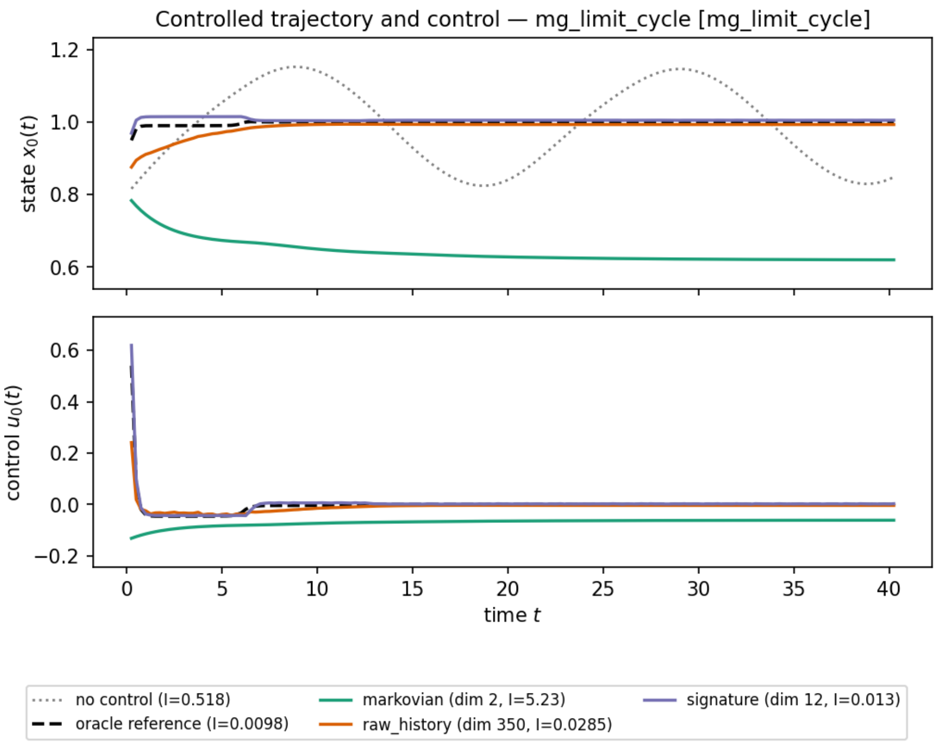

Figure 6: `mg_limit_cycle` (canonical off-sheet budget, $N_{\text{off}} = 500$). A clean H_1 picture: the markovian critic (green) cannot represent the value ($R^2 = 0.14$) and drives the state to a *wrong* equilibrium near $x_0 = 0.62$ (cost $J = 5.23$), while the history window (orange) and signature (purple) both regulate to $x^* = 1$. At this off-sheet budget the signature leads ($J = 0.013$ versus the history window's 0.029); Finding 5.5 shows that scaling the off-sheet data to off/dim = 4 reverses the ordering.

Provenance caveats. The Hopfield/Mackey–Glass sweeps and the effective-rank measurement are single-seed (the operative quantity there is the data/conditioning axis, not seed variance); the canonical H_1/H_2 table is five-seed. The Mackey–Glass off-sheet curve is now complete for both examples (the last mg_chaotic point, off/dim = 4, at $\sim 10^4$ Pontryagin solves, finished after the first draft and is included). The trajectory figures are local regenerations whose well-conditioned costs match the cluster `summary.json` to four decimals; the ill-conditioned signature on the platoon reproduces only to ∼10% (local 0.140 versus cluster 0.164), itself a symptom of the condition number of Figure 3.